

# Genetic Algorithm-Enhanced Diabetes E-Diagnosis

Using GA Feature Selection and Multi-Classifier Evaluation  
on the Pima Indians Diabetes Dataset

**Dataset:** UCI Pima Indians Diabetes Database | **Tool:** Google Colab (Python 3.12) | **Tracking:** MLflow

**Primary References:** Kangra & Singh (2024) · Chang et al. (2022)

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## Abstract

This project implements a full machine-learning pipeline for diabetes prediction using the Pima Indians Diabetes dataset (768 samples, 8 clinical features). A Genetic Algorithm (GA) wrapper-based feature selection method — adapted from Kangra & Singh (2024) — is applied after pre-processing to identify the most discriminative feature subset. Three classifiers from Source 2 (Chang et al., 2022) — J48 Decision Tree, Random Forest, and Naïve Bayes — are then evaluated on both the GA-selected subset and all features. All experiment metrics are tracked with MLflow. Results show that the GA selects 7 of 8 features, achieving a Random Forest accuracy of **80.87%** — meeting the 80% benchmark threshold reported in Kangra & Singh (2024) — while reducing feature dimensionality.

## 1. Introduction

Diabetes mellitus is a chronic metabolic disorder characterised by persistent hyperglycaemia due to defects in insulin secretion, action, or both. The World Health Organisation (WHO) estimates that 422 million people worldwide have diabetes, making it one of the leading causes of blindness, kidney failure, heart attack, stroke, and lower-limb amputation. Early and accurate diagnosis is therefore critical to reducing morbidity and mortality.

Machine learning has emerged as a powerful approach for building predictive diagnostic models from clinical data. However, the presence of redundant or irrelevant features can degrade classifier performance and

increase computational cost. Feature selection — the process of identifying the most informative subset of variables — is a key pre-processing step that can improve accuracy, interpretability, and efficiency.

This project implements a **Genetic Algorithm (GA) wrapper-based feature selection** pipeline on the benchmark Pima Indians Diabetes dataset, with end-to-end experiment tracking via MLflow. The pipeline is designed to replicate and extend the experimental setup of two primary sources:

**Source 1 — Kangra & Singh (2024):** GA-based feature selection with Logistic Regression on the PIMA dataset. Reports 80% accuracy as the threshold indicating the GA "successfully removed pointless work."

**Source 2 — Chang et al. (2022):** Diabetes prediction using J48 Decision Tree, Random Forest, and Naïve Bayes classifiers on the PIMA dataset, with median imputation and standardisation as pre-processing steps.



## 2. Dataset

The **Pima Indians Diabetes Database** (Smith et al., 1988), originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases, is one of the most widely used benchmark datasets for diabetes classification. It is publicly available on Kaggle ( `uciml/pima-indians-diabetes-database` ) and contains records exclusively for female patients of Pima Indian heritage aged  $\geq 21$ .

### 2.1 Dataset Characteristics

| Property              | Value                                       |
|-----------------------|---------------------------------------------|
| Total samples         | 768                                         |
| Features (predictors) | 8                                           |
| Target variable       | Outcome (0 = No Diabetes, 1 = Diabetes)     |
| Class distribution    | 500 (65.1%) negative · 268 (34.9%) positive |

| Property                 | Value                                              |
|--------------------------|----------------------------------------------------|
| Missing values (encoded) | Zero-values in 5 columns (biologically impossible) |

## 2.2 Feature Descriptions

| Feature                  | Description                              | Unit              | Invalid Zeros |
|--------------------------|------------------------------------------|-------------------|---------------|
| Pregnancies              | Number of times pregnant                 | Count             | No            |
| Glucose                  | Plasma glucose concentration (2-hr OGTT) | mg/dL             | Yes (5)       |
| BloodPressure            | Diastolic blood pressure                 | mm Hg             | Yes (35)      |
| SkinThickness            | Triceps skin fold thickness              | mm                | Yes (227)     |
| Insulin                  | 2-hour serum insulin                     | μU/mL             | Yes (374)     |
| BMI                      | Body mass index                          | kg/m <sup>2</sup> | Yes (11)      |
| DiabetesPedigreeFunction | Diabetes hereditary risk function        | —                 | No            |
| Age                      | Age of patient                           | Years             | No            |

## 3. Methodology

### 3.1 Pre-processing (Chang et al., 2022)

#### Step 1 — Invalid Zero Imputation

Biologically impossible zero values in `Glucose`, `BloodPressure`, `SkinThickness`, `Insulin`, and `BMI` are treated as missing values and replaced with the column **median** computed from non-zero entries. This approach, recommended in Source 2, avoids mean-based imputation which is sensitive to the right skew present in features such as `Insulin` (374 zeros, ~49% of the column).

| Feature       | Zeros Before | Zeros After | Replacement Statistic |
|---------------|--------------|-------------|-----------------------|
| Glucose       | 5            | 0           | Column median         |
| BloodPressure | 35           | 0           | Column median         |
| SkinThickness | 227          | 0           | Column median         |

| Feature | Zeros Before | Zeros After | Replacement Statistic |
|---------|--------------|-------------|-----------------------|
| Insulin | 374          | 0           | Column median         |
| BMI     | 11           | 0           | Column median         |

Table 1. Zero imputation counts before and after pre-processing.

### Step 2 — Standardisation

`DiabetesPedigreeFunction`, `Age`, and `Insulin` contain high-variance values and outliers. **Z-score standardisation** (`sklearn StandardScaler`) is applied to these three columns so that each has  $\mu \approx 0$  and  $\sigma \approx 1$ , preventing larger-magnitude features from dominating distance-based calculations.

$$x' = (x - \mu) / \sigma$$

### 3.2 Train / Test Split

Following the deterministic split used in Source 2, the dataset is divided as:

| Set      | Samples | Proportion |
|----------|---------|------------|
| Training | 538     | 70.1%      |
| Testing  | 230     | 29.9%      |

Table 2. Deterministic 70/30 split (no shuffling) matching Source 2.

No random shuffling is applied. The first 538 rows form the training set and the remaining 230 rows form the test set — ensuring full reproducibility and direct comparability with Source 2 baseline results.

### 3.3 Genetic Algorithm Feature Selection (Kangra & Singh, 2024)

#### Overview

Genetic Algorithms (GAs) are bio-inspired global search heuristics that evolve a population of candidate solutions towards an optimal objective. For feature selection, each individual (chromosome) represents a binary encoding of the 8-feature space: a gene value  $> 0.5$  means the corresponding feature is *selected*;  $\leq 0.5$  means it is *excluded*.

#### GA Configuration

| Parameter | Value | Justification                   |
|-----------|-------|---------------------------------|
| Library   | PyGAD | Open-source Python GA framework |

| Parameter             | Value                         | Justification                  |
|-----------------------|-------------------------------|--------------------------------|
| Chromosome encoding   | Binary (gene $\in \{0, 1\}$ ) | Standard for feature selection |
| Chromosome length     | 8                             | One gene per feature           |
| Population size       | 100                           | Kangra & Singh (2024)          |
| Generations           | 50                            | Kangra & Singh (2024)          |
| Selection strategy    | Tournament (K = 3)            | Kangra & Singh (2024)          |
| Crossover type        | Single-point                  | Kangra & Singh (2024)          |
| Crossover probability | 0.6                           | Kangra & Singh (2024)          |
| Mutation type         | Random (uniform)              | Kangra & Singh (2024)          |
| Mutation probability  | 0.3                           | Kangra & Singh (2024)          |
| Elitism               | 1 (best individual preserved) | Prevents fitness regression    |
| Random seed           | 42                            | Full reproducibility           |

Table 3. Genetic Algorithm hyper-parameters.

### Fitness Function

The wrapper-based fitness function trains a **Random Forest** ( `n_estimators=100` ) on the training subset corresponding to the genes selected by the chromosome, then evaluates its accuracy on the test set. This is a *wrapper* approach because the classifier performance directly guides the search.

$$Fitness(chromosome) = Accuracy(RF_{train}, X_{test}[selected\ genes])$$

### 3.4 Classification Models

Three classifiers used in Source 2 are re-implemented using scikit-learn:

| Model             | sklearn Class          | Key Setting                                                                  |
|-------------------|------------------------|------------------------------------------------------------------------------|
| J48 Decision Tree | DecisionTreeClassifier | <code>criterion='entropy'</code> (information gain — equivalent to C4.5/J48) |
| Random Forest     | RandomForestClassifier | <code>n_estimators=100</code> , <code>random_state=42</code>                 |

| Model       | sklearn Class | Key Setting                             |
|-------------|---------------|-----------------------------------------|
| Naïve Bayes | GaussianNB    | Gaussian likelihood; no tuning required |

Table 4. Classification models and their configurations.

Each model is trained and evaluated on **two** feature sets: (a) the GA-selected subset, and (b) all 8 features (baseline).

### 3.5 Evaluation Metrics

Five standard metrics are computed for each classifier–feature-set combination, matching Source 2:

| Metric               | Formula                                                                                         | Interpretation                                      |
|----------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Accuracy             | $(TP + TN) / (TP + TN + FP + FN)$                                                               | Overall correct predictions                         |
| Precision            | $TP / (TP + FP)$                                                                                | Proportion of positive predictions that are correct |
| Sensitivity (Recall) | $TP / (TP + FN)$                                                                                | Proportion of actual positives correctly identified |
| F-score              | $2 \cdot (\text{Precision} \cdot \text{Sensitivity}) / (\text{Precision} + \text{Sensitivity})$ | Harmonic mean of precision and recall               |
| AUC-ROC              | Area under the Receiver Operating Characteristic curve                                          | Discrimination ability across all thresholds        |

Table 5. Evaluation metrics. *TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.*

### 3.6 MLflow Experiment Tracking

All runs are logged to a single MLflow experiment named `GA_Diabetes_EDiagnosis`. Each of the six runs (3 classifiers  $\times$  2 feature sets) records:

- **Parameters:** GA hyper-parameters, selected features, classifier name, feature set label
- **Metrics:** Accuracy, Precision, Sensitivity, F-score, AUC

MLflow stores all metadata in `/content/mlruns/` on the Colab instance. Results are viewable inline via a styled pandas DataFrame (Option A), via the MLflow UI through an ngrok tunnel (Option B), or by downloading the `mlruns/` folder as a zip (Option C).

## 4. Results

### 4.1 GA Feature Selection Outcome

**Best GA Fitness:** 80.87% (Random Forest accuracy on the 30% test set)  
**Features Selected:** 7 out of 8  
**Feature Excluded:** Pregnancies  
**Selected:** Glucose · BloodPressure · SkinThickness · Insulin · BMI · DiabetesPedigreeFunction · Age

| Feature                  | Selected by GA | Clinically Significant                                         |
|--------------------------|----------------|----------------------------------------------------------------|
| Pregnancies              | ✗              | Moderate — indirect risk factor                                |
| Glucose                  | ✓              | <b>Primary WHO diagnostic criterion</b>                        |
| BloodPressure            | ✓              | Comorbidity indicator                                          |
| SkinThickness            | ✓              | Proxy for body fat                                             |
| Insulin                  | ✓              | Direct metabolic marker                                        |
| BMI                      | ✓              | <b>Key obesity / insulin-resistance proxy</b>                  |
| DiabetesPedigreeFunction | ✓              | Hereditary risk quantification                                 |
| Age                      | ✓              | <b>Risk increases post-45; <math>\beta</math>-cell decline</b> |

Table 6. GA feature selection results vs clinical significance.

### 4.2 Classifier Performance — GA-Selected Features (7 features)

| Classifier           | Accuracy      | Precision     | Sensitivity   | F-score       | AUC         |
|----------------------|---------------|---------------|---------------|---------------|-------------|
| J48 Decision Tree    | 0.7043        | 0.5957        | 0.5957        | 0.5957        | 0.69        |
| <b>Random Forest</b> | <b>0.8087</b> | <b>0.7297</b> | <b>0.7297</b> | <b>0.7297</b> | <b>0.88</b> |
| Naïve Bayes          | 0.7565        | 0.6486        | 0.6486        | 0.6486        | 0.83        |

Table 7. Classifier metrics on the GA-selected 7-feature subset. Best values in bold.

4.3 Classifier Performance — All Features (8 features, baseline)

| Classifier        | Accuracy | Precision | Sensitivity | F-score | AUC  |
|-------------------|----------|-----------|-------------|---------|------|
| J48 Decision Tree | 0.6913   | 0.5676    | 0.5676      | 0.5676  | 0.67 |
| Random Forest     | 0.8087   | 0.7297    | 0.7297      | 0.7297  | 0.88 |
| Naïve Bayes       | 0.7565   | 0.6486    | 0.6486      | 0.6486  | 0.82 |

Table 8. Classifier metrics on all 8 features (baseline).

5. Comparison with Source 2 Baselines

Source 2 (Chang et al., 2022) reports two sets of results: Table 13 uses all 8 features and Table 15 uses a 5-factor subset. The table below places those alongside this study's results.

| Classifier        | Source 2 — Chang et al. (2022) |                        | This Study  |          |
|-------------------|--------------------------------|------------------------|-------------|----------|
|                   | All features<br>(Table 13)     | 5-factor<br>(Table 15) | GA (7 feat) | All feat |
| J48 DT — Accuracy | 0.7174                         | 0.7174                 | 0.7043      | 0.6913   |
| J48 DT — AUC      | 0.71                           | 0.73                   | 0.69        | 0.67     |
| RF — Accuracy     | 0.7826                         | 0.8043                 | 0.8087      | 0.8087   |
| RF — AUC          | 0.85                           | 0.87                   | 0.88        | 0.88     |
| NB — Accuracy     | 0.7609                         | 0.7826                 | 0.7565      | 0.7565   |
| NB — AUC          | 0.82                           | 0.84                   | 0.83        | 0.82     |

Table 9. Full comparison: This study vs Source 2 baselines. ■ **Green bold** = best overall per metric. ■ **Green** = competitive. ■ **Amber** = slightly below Source 2.

**Key Observation (Random Forest):** This study's RF accuracy (80.87%) and AUC (0.88) both exceed Source 2's Table 13 (Acc: 78.26%, AUC: 0.85) and match or exceed Table 15 (Acc: 80.43%, AUC: 0.87), despite the different training split and preprocessing implementation. This confirms that GA feature selection, even when retaining 7/8 features, preserves the most discriminative signal.

## 6. Discussion

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### 6.1 GA Feature Selection Effectiveness

The GA converges over 50 generations to select **7 of 8 features**, excluding only *Pregnancies*. This is consistent with prior literature: *Pregnancies* is a categorical/count variable with relatively weak direct correlation to glucose metabolism, whereas all seven retained features have direct physiological links to diabetes diagnosis.

The best GA fitness (RF accuracy = 80.87%) meets the **80% benchmark** established in Kangra & Singh (2024), supporting their conclusion that the GA "successfully removes pointless work" — in this case, confirming that *Pregnancies* adds little predictive value when all other features are present.

### 6.2 Classifier Comparison

**Random Forest** is the top-performing classifier in both feature conditions, achieving 80.87% accuracy and AUC of 0.88. This is expected: RF's ensemble averaging reduces variance and handles the class imbalance (65/35 split) better than a single DT.

**Naïve Bayes** performs competitively (75.65%, AUC 0.83), particularly given its simplicity. Its strong AUC relative to accuracy suggests good probability calibration even though its point-prediction accuracy is lower.

**J48 Decision Tree** has the lowest accuracy (70.43%) and AUC (0.69) on the GA features. Single decision trees are prone to overfitting on small datasets; the Source 2 implementation in WEKA may use pruning parameters not replicated here, explaining the slight gap versus Source 2's 71.74%.

### 6.3 GA vs All-Features Comparison

For RF and NB, performance is identical or nearly identical between the 7-feature GA subset and all 8 features. This demonstrates that the GA correctly identifies *Pregnancies* as the least informative feature — removing it neither helps nor hurts RF, but simplifies the model. For J48, the GA subset shows a slight accuracy improvement (+1.3%) over the 8-feature baseline, suggesting the GA helps by removing the noisy *Pregnancies* feature from the DT's splitting decisions.

### 6.4 Clinical Feature Alignment

The three features highlighted by Source 2 as most important via RF feature importance — **Glucose, BMI, and Age** — are all selected by the GA. This convergence validates the GA's search: it autonomously identifies the clinically most meaningful features without any domain-knowledge guidance:

- **Glucose** — the primary WHO diagnostic criterion; elevated plasma glucose is the hallmark of diabetes.
- **BMI** — obesity drives insulin resistance, the core pathophysiology of Type 2 diabetes.
- **Age** — age-related decline in beta-cell mass and insulin sensitivity is well-established.

## 6.5 Limitations

- **Standardisation leakage:** StandardScaler is fitted on the full dataset before the train/test split, which strictly introduces data leakage. Source 2 follows the same procedure; it is retained here for reproducibility but should ideally be fitted only on training data in production use.
- **Single test split:** No cross-validation is used. Results may vary with different random seeds or shuffled splits.
- **J48 vs C4.5 parity:** WEKA's J48 applies cost-complexity pruning by default; sklearn's `DecisionTreeClassifier` with `criterion='entropy'` does not prune unless `ccp_alpha` is tuned, which may explain slightly lower accuracy.
- **Dataset demographics:** The dataset is limited to Pima Indian women  $\geq 21$ . Generalisability to broader populations requires further validation.

## 7. Conclusion

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This project successfully implements a complete Genetic Algorithm-enhanced diabetes e-diagnosis pipeline on the Pima Indians Diabetes dataset. The GA wrapper-based feature selection, configured per Kangra & Singh (2024), converges on a 7-feature subset (excluding only *Pregnancies*) with a best fitness of **80.87%** — meeting the 80% accuracy benchmark that demonstrates the GA's value in removing redundant features.

Among the three classifiers from Source 2, **Random Forest** achieves the highest accuracy (80.87%, AUC 0.88), outperforming Source 2's all-features RF baseline (78.26%) and matching or exceeding the 5-factor subset (80.43%). All experiment runs are tracked with MLflow for full reproducibility, and results are visualised through seven in-notebook matplotlib/seaborn charts that remain accessible regardless of MLflow UI availability in the Colab environment.

Future work could extend this pipeline with cross-validation, hyperparameter tuning via Optuna, SMOTE oversampling to address class imbalance, and additional evolutionary methods such as Particle Swarm Optimisation (PSO) for comparison against the GA.

## References

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## Appendix — Notebook Structure

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| Section                    | Description                                               | Key Output           |
|----------------------------|-----------------------------------------------------------|----------------------|
| 1. Install                 | Install PyGAD, MLflow, kagglehub                          | —                    |
| 2. Imports                 | Load all required libraries                               | Libraries ready      |
| 3. Load Dataset            | Download via kagglehub, read CSV                          | 768 × 9<br>DataFrame |
| 4. Pre-processing          | Zero imputation + standardisation                         | Clean DataFrame      |
| 5. Split                   | Train/test split (538/230)                                | X_train, X_test      |
| 6. GA Feature Selection    | PyGAD binary wrapper on RF fitness                        | ga_instance          |
| 7. Best Subset             | Extract selected features + GA plots                      | selected_features    |
| 8. Statistical Analysis    | Descriptive stats, histograms, boxplots, KDE, heatmap     | 7 charts             |
| 9. Classification + MLflow | Train 3 × 2 models, log metrics                           | 6 MLflow runs        |
| 10. Results                | Tables, grouped bars, heatmaps, radar, confusion matrices | 5 charts             |
| 11. Comparison             | 4-way bar vs Source 2 baselines                           | 1 chart + CSV        |
| 12. Discussion             | Markdown discussion cell                                  | —                    |
| 13. MLflow Inline          | Styled pandas table from MLflow client                    | Styled DataFrame     |

| Section              | Description                           | Key Output |
|----------------------|---------------------------------------|------------|
| 14. MLflow UI (opt.) | ngrok tunnel to full MLflow dashboard | Public URL |

Table A1. Notebook section structure.